

Team 94321C Engineering Notebook^{*Format later*}

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Robot Design

Our initial robot design is heavily inspired by this year's hero bot, Flop. Flop has showed remarkable scoring abilities by flipping bean bags from front to back and landing them on Level 3 with consistency. However, despite its effectiveness, Flop has some limitations, most importantly in speed. The robot can sometimes operate slower than desired, and the motor may overheat.

To fix these challenges and improve our robot's performance, we made several changes. One of the most important changes was moving the intake motor closer to the robot's pivot point. This change reduces the motor's effort, meaning a smoother and more efficient flipping process.

In addition, we added small "carrots" in front of the robot. These attachments make it easier to drive over the loading zone. These carrots are small, triangular pieces that have smooth edges. Overall, these changes are intended to create a faster and effective robot that can perform consistently under a variety of conditions.

Our second robot is not a robot that flips beanbags onto L3 and L4. It uses a linear launcher powered by a ratchet to launch the beanbags onto Level 3 and Level 4. It has a string to pull and load the launcher, and uses rubber bands to simply launch to beanbag. To prevent the launcher from flying off the linear slides, we put a couple of 1 by 1 beams to stop it. This launcher design makes it smaller and more compact. It also makes it a little bit easier to tune. However, this specific robot design has some issues, such as a part sometimes flying off or the ratchet getting loose.

To switch between L3 and L4, our robot has pneumatics that adjusts the angle of the launcher. However, this means we are using the same launcher for L3 and L4, so we have to make it work for both. Our air tank is on the side because it fits there pretty well, and our pneumatics are all on one side. We only used one side of the solenoid to work for both cylinders because that makes the code and wiring a lot simpler.

Our intake is similar to the intake of the hero bot, Flop, but instead of button the intake directly on the launcher, which would make it less stable and harder to intake, we put it on the back of the robot. When we drive into the beanbag, the driver presses a button and it intakes onto a temporary storage for one beanbag. Then, it flips the intake over onto the loaded launcher, and spins the intake until it rests onto the launcher. This is fast, but also has some issues, such as only being able to intake the beanbag one way or having all the weight in the beanbag shifted to one side of the beanbag.

2026-2027 Level Up Game Explanation

In the 2026-2027 VEX IQ game, Level Up, teams will be challenged to score beanbags by placing them on a pyramid structure or L4 goal. The beanbags are 3" x 4" and easy to manipulate but require precision placement. The pyramid has three different levels, each worth a different number of points. Both teams get 3 points for the first level, 6 points for the second level, and 12 points for the third level. For our first competition, we are aiming for Level 3, as our current robot is not reliable enough to consistently reach Level 4 but is able to score at Level 3.

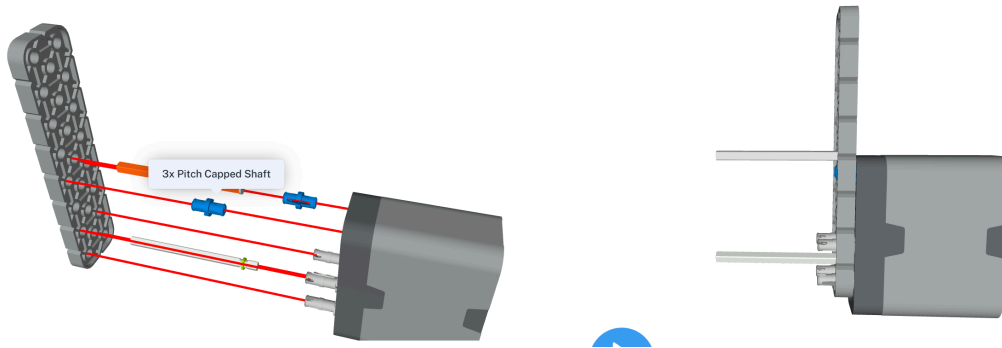
The pyramid's Level 3 stands 8 inches tall, presenting a challenge for our robot. Our robot must be less than 6.25 inches tall when folded to be able to successfully move under the barriers. Also, the ramp leading up to the pyramid has a height of 0.75 inches, which means the wheels cannot just be centered normally on the robot. In order for our robot to score, it must go about 66 inches from the loading zone to the scoring zone.

For this year's game, the robot starts on the opposite side of the loading zone, which means that the teams will be loading for each other instead of their own robot. This means that good communication is very important because you need to know when to load, how to put the beanbag, and what should be avoided.

5/23/2026

Today, we finished the assembly of the hero bot Flop, which was the first step in our project. But during the process, we found a problem with one of the shafts in the instructions. A 3-pitch shaft was incorrectly labeled as a 4-pitch capped shaft, which could lead to issues in future references and operations.

You can see in the pictures that the 4-pitch capped shaft is clearly labeled as a 3-pitch capped shaft. This mislabeling is critical to fix because it can affect the bot's performance and functionality. The images also show a green motor shaft of similar length to the incorrectly labeled shaft. Important to keep in mind that green motor shafts are always 4-pitch, which reinforces the importance of accurate labeling to prevent any possible operational errors.



Second Robot

Our second prototype robot, which we named the Hippomech, was a test after we watched a few videos for inspiration for the robot. It was a test launcher we started working on in June. We are not taking it to our first competition because it is not very consistent and we don't have a lot of driving practice with it.

The Hippomech consists of five parts. The first part is the drivetrain. It's a 2 motor drive that connects the wheels with chains. This design enables the robot to drive using only two motors while still powering all four. The motors are on the inside of the robot, and flat down because that saves space, since there isn't much space on the outside.

The second section is the launcher. It is a linear launcher, and the beanbag goes on the flat part that uses a string to load the and pull down the beanbag. The two 2x2 connectors prevent the beanbags from falling off. The linear slides allow smooth movement, and the 1x1 beam at the front prevents the launcher from falling off. There are two rubber bands that pull the launcher so that there is enough tension.

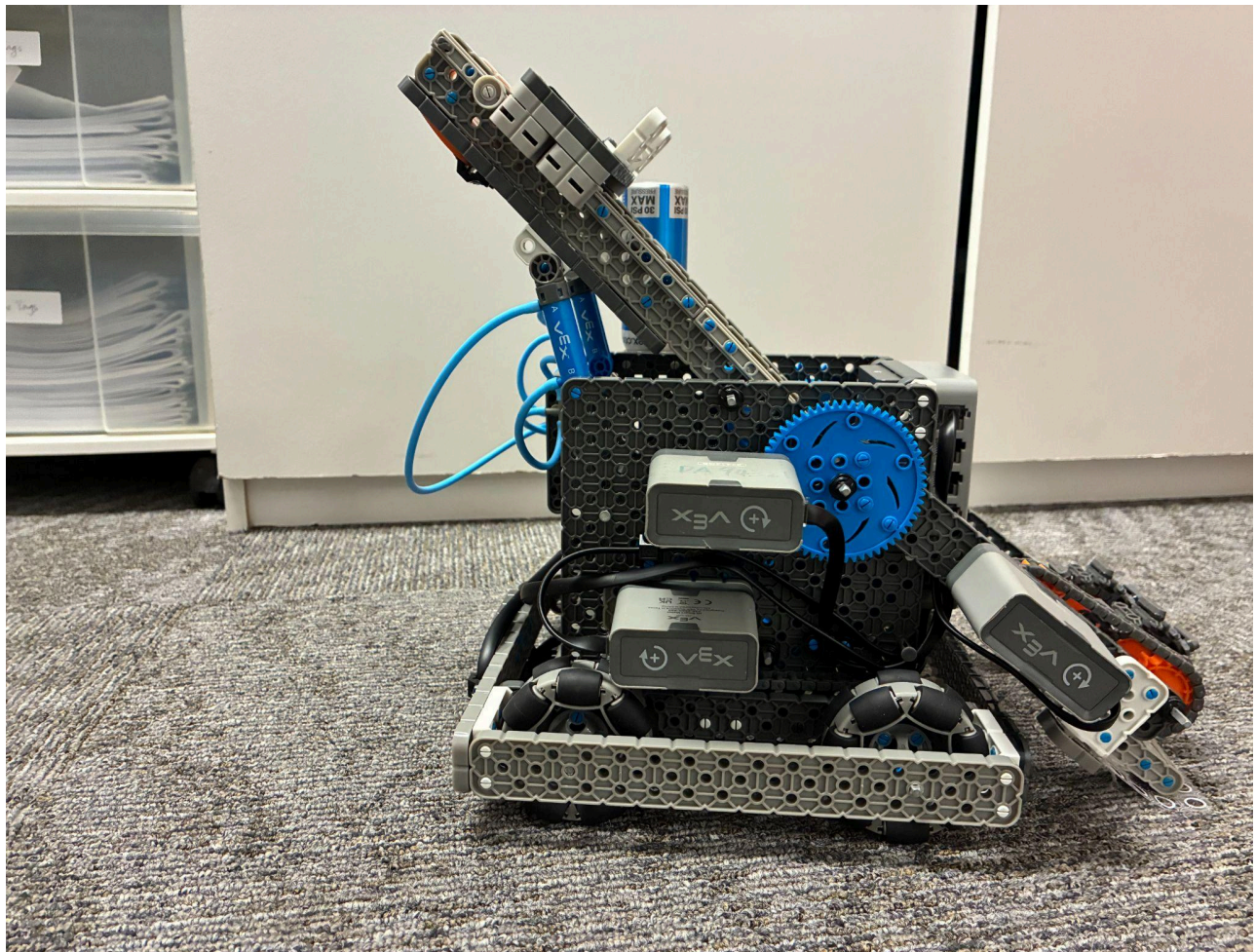
The third section is the area below the launcher. There are 2 pneumatic pistons that can extend or retract to change the angle it is launching from. When extended, the angle is good for Level 4 and when retracted, it is perfect for Level 3.

The fourth section is the intake, and the way it works is it flips the beanbag onto the loaded launcher. We chose to flip it on while it is already loaded because going to the beanbag and grabbing it takes time, which we

can use to load the launcher. Then, the intake flips over and puts the beanbag onto the launcher.

The fifth and final section is the center, which has most of our motors, our pneumatics, and other wiring. This makes everything tidier and makes it less likely to clog on gears and chains, or other moving parts. We learned this last year, where our first robot was messy, so it made it really hard to get consistent runs. One drawback to doing something like this is it makes it much harder to fix the pneumatics and wiring. However, if we make it tight the first time we do it, we don't have to redo it. This year, we aren't using as many pneumatics as last year.

Video link: <https://www.youtube.com/watch?v=2XUrQIFFe0Q>



Third Robot

Our third robot is an improvement of the hippomech. Because the hippomech was not that accurate or fast, we built Strawberry to solve those problems. Strawberry has a more accurate launcher because the launcher is a curved one (not a linear launcher) and it has bumpers on the back so it knows when it is fully loaded.

Strawberry also has various aligners, such as a beam on the right side to align into the ramp and two “crab arms” on the top to align into the top bar. We also have a double reverse 2 bar for Level 4 and the intake.